

	Yasin Rohani s330058
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Terminato	16 febbraio 2026, 15:51
Tempo impiegato	1 ora 40 min.
Valutazione	12,53 su un massimo di 32,00 (39%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17

Domanda 1

Completo

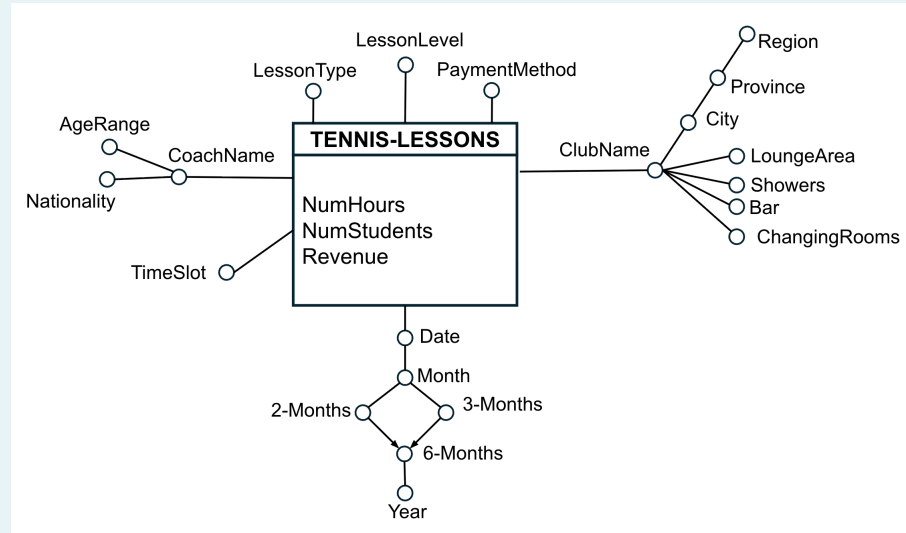
Punteggio
ottenuto 2,25
su 4,00

Extended SQL (4 points)

The following data warehouse stores information related to the management of tennis lessons for a chain of tennis clubs located throughout Italy. We want to analyze, on a daily basis and by time slot (morning from 9:00 a.m. to 1:00 p.m., afternoon from 1:00 p.m. to 5:00 p.m., and evening from 5:00 p.m. to 9:00 p.m.), the number of hours, the number of students, and the amount of revenue collected.

The tennis clubs are characterized by a geographical hierarchy and by the available services, such as the presence of a lounge area, bar, showers, and changing rooms (configuration attribute). Tennis coaches are characterized by nationality and age range. The analysis also considers the lesson type (individual or group), the lesson level (beginner/advanced/competition), and the payment method (cash/card).

The data warehouse is characterized by the following conceptual schema and the corresponding logical schema.



TENNIS-CLUB (IDClub, ClubName, City, Province, Region, LoungeArea, Showers, Bar, ChangingRooms)

COACH (IDCoach, CoachName, Nationality, AgeRange)

TIME-DATE (IDTimeD, Date, Month, 2-Months, 3-Months, 6-Months, Year)

TIME-HOUR (IDTimeH, TimeSlot)

JUNK-LESSON (IDJL, LessonType, LessonLevel, PaymentMethod)

TENNIS-LESSONS (IDClub, IDCoach, IDTimeD, IDTimeH, IDJL, NumHours, NumStudents, Revenue)

For each tennis-club that has showers and a lounge area, and for each two-month period, show:

- average number of lesson hours per student and average revenue per lesson
- ratio between the tennis club's lesson hours and the total lesson hours of all tennis clubs located in the same region
- cumulative number of lesson hours over successive two-month periods, separately by year

Perform the analyses for each lesson type.

```
SELECT ClubName 2-Month
SUM(NumHours)/SUM(NumStudents), SUM(Revenue) / SUM(NumHours)
SUM(NumHours)/SUM(SUM(NumHours)OVER(PARTION BY Region))
SUM(SUM(NumHours))OVER(PARTION BY Year Region
                        ORDER BY 2-Month
                        ROWS UNBOUNDED PRECEDING)
FROM TENNIS-LESSON T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI
WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL

GROUP BY ClubName Shower Bar 2-Month Year Region , LessonType
```

Commento:

```
SELECT ClubName 2-Month
SUM(NumHours)/SUM(NumStudents), SUM(Revenue) / SUM(NumHours)
SUM(NumHours)/SUM(SUM(NumHours)OVER(PARTION BY Region, LessonType, 2M))
SUM(SUM(NumHours))OVER(PARTION BY Year Region ClubName, LessonType
                        ORDER BY 2-Month
                        ROWS UNBOUNDED PRECEDING)
FROM TENNIS-LESSON T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI
WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL AND Showers='yes' AND
LoungeArea ='yes'
```

Domanda 2

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

Group by anticipation (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
DOCTOR (DID, FirstName, LastName, Department, Salary)
PROTOCOL (PID, ProtocolName, TherapyType, MedicalArea, Duration)
EXECUTION (DID, PID, ExecutionDate)
PATIENT (PTID, FirstName, LastName, BirthDate, ResidenceCity)
ASSIGNMENT (PTID, PID, Outcome, AssignmentDate)
```

Assume the following cardinalities:

- $\text{card}(\text{DOCTOR}) = 2 \cdot 10^3$ tuples
 - Distinct values of the Department = 10
- $\text{card}(\text{PROTOCOL}) = 2 \cdot 10^4$ tuples
 - Distinct values of MedicalArea = 20
- $\text{card}(\text{EXECUTION}) = 4 \cdot 10^5$ tuples
 - $\text{MIN}(\text{ExecutionDate}) = 1/1/2006$, $\text{MAX}(\text{ExecutionDate}) = 31/12/2025$
- $\text{card}(\text{PATIENT}) = 10^5$ tuples
 - Distinct values of ResidenceCity = 200
 - $\text{MIN}(\text{BirthDate}) = 1/1/1950$, $\text{MAX}(\text{BirthDate}) = 31/12/2009$
- $\text{card}(\text{ASSIGNMENT}) = 4 \cdot 10^5$ tuple

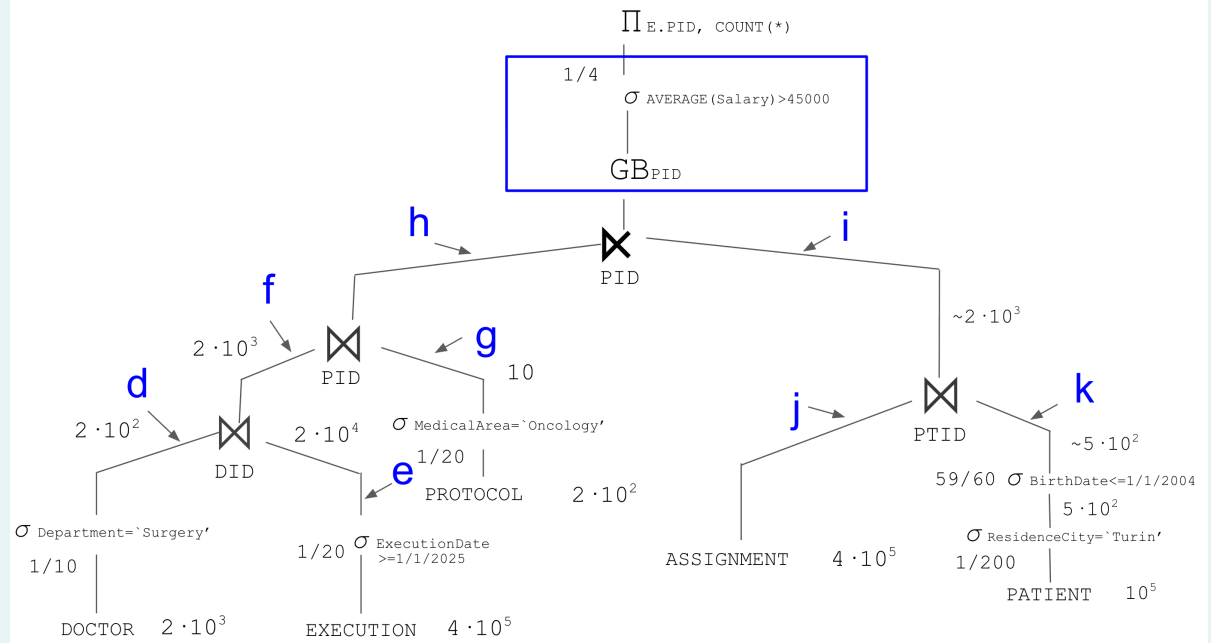
Furthermore, assume the following reduction factor for the having clauses:

$\text{HAVING AVERAGE}(\text{D.Salary}) > 45000 = 1/4$

Consider the following query:

```
SELECT E.PID, COUNT(*)
FROM DOCTOR D, PROTOCOL P, EXECUTION E
WHERE D.DID = E.DID
      AND P.PID = E.PID
      AND P.MedicalArea = 'Oncology'
      AND E.ExecutionDate >= 1/1/2025
      AND D.Department = 'Surgery'
      AND E.PID IN (
        SELECT A.PID
        FROM ASSIGNMENT A, PATIENT PT
        WHERE A.PTID = PT.PTID
              AND PT.ResidenceCity = 'Turin'
              AND PT.BirthDate <= 1/1/2009
      )
GROUP BY E.PID
HAVING AVERAGE(D.Salary) > 45000;
```

The figure below represents the query tree for the query above.



Analyze the anticipation of the GROUP BY E.PID HAVING AVERAGE(D.Salary) > 45000 represented in the box. Select the solution that allows for the maximum efficiency in executing the query (if it exists).

- ☐ a. It is possible to anticipate it in branch d
- ☐ b. It is possible to anticipate it in branch g
- ☐ c. It is possible to anticipate it in branch e
- ☐ d. It is possible to anticipate it in branch h
- ☐ e. It is possible to anticipate it in branch j
- ☐ f. It is possible to anticipate it in branch k
- ☒ g. It is possible to anticipate it in branch f ✓
- ☐ h. It is not possible to anticipate the GROUP BY E.PID HAVING AVERAGE(D.Salary) > 45000
- ☐ i. It is possible to anticipate it in branch i

Risposta corretta.

La risposta corretta è: It is possible to anticipate it in branch f

Domanda 3

Risposta
errata

Punteggio
ottenuto -0,23
su 1,50

1.5 points, -15% penalty for each wrong answer

Notation

RN(x): read of object x by transaction T_n

WN(x): write of object x by transaction T_n

Exercise text

Consider the following schedule S of four transactions:

S = W0(a), R1(a), W1(b), R2(b), W2(c), R3(c), R0(d), W1(d), W3(e), W0(f)

Verify whether schedule S is conflict-serializable. If so, select the equivalent serial schedule

- ☐ a. W0(a), R0(d), W0(f), R1(a), W1(b), W1(d), R2(b), W2(c), R3(c), W3(e)
- ☒ b. W0(a), R0(d), W0(f), R1(a), W1(c), W1(d), R2(b), W2(c), R3(c), W3(e) ✗
- ☐ c. R1(a), W1(b), R2(b), W2(c), R3(c), W3(e), W0(a), R0(d), W1(d), W0(f)

- ☐ d. W0(a), R1(a), W1(b), R2(b), W2(c), R3(c), W1(d), R0(d), W3(e), W0(f)
- ☐ e. S is not conflict-serializable
- ☐ f. W0(a), R1(a), W1(b), W1(d), R2(b), W2(c), R3(c), R0(d), W3(e), W0(f)

Risposta errata.

La risposta corretta è:

W0(a), R0(d), W0(f), R1(a), W1(b), W1(d), R2(b), W2(c), R3(c), W3(e)

Domanda 4

Completo

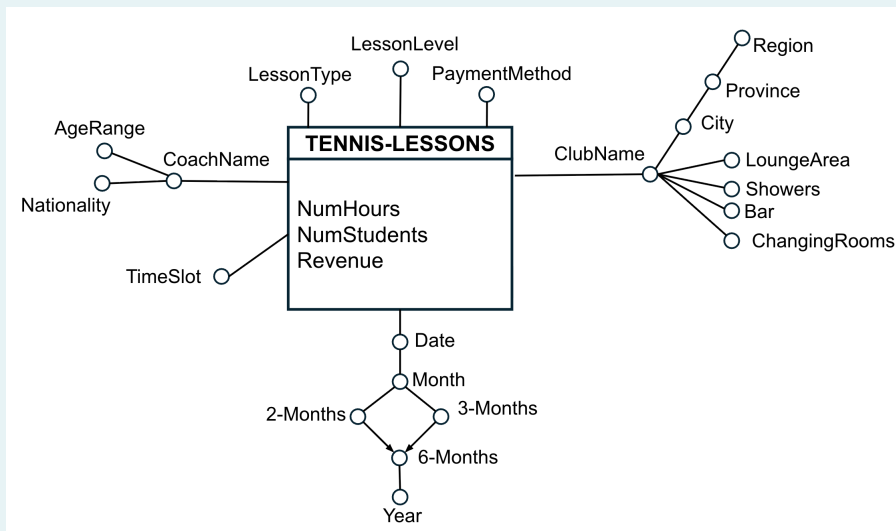
Punteggio
ottenuto 2,25
su 3,00

Extended SQL (3 points)

The following data warehouse stores information related to the management of tennis lessons for a chain of tennis clubs located throughout Italy. We want to analyze, on a daily basis and by time slot (morning from 9:00 a.m. to 1:00 p.m., afternoon from 1:00 p.m. to 5:00 p.m., and evening from 5:00 p.m. to 9:00 p.m.), the number of hours, the number of students, and the amount of revenue collected.

The tennis clubs are characterized by a geographical hierarchy and by the available services, such as the presence of a lounge area, bar, showers, and changing rooms (configuration attribute). Tennis coaches are characterized by nationality and age range. The analysis also considers the lesson type (individual or group), the lesson level (beginner/advanced/competition), and the payment method (cash/card).

The data warehouse is characterized by the following conceptual schema and the corresponding logical schema.



TENNIS-CLUB (IDClub, ClubName, City, Province, Region, LoungeArea, Showers, Bar, ChangingRooms)

COACH (IDCoach, CoachName, Nationality, AgeRange)

TIME-DATE (IDTimeD, Date, Month, 2-Months, 3-Months, 6-Months, Year)

TIME-HOUR (IDTimeH, TimeSlot)

JUNK-LESSON (IDJL, LessonType, LessonLevel, PaymentMethod)

TENNIS-LESSONS (IDClub, IDCoach, IDTimeD, IDTimeH, IDJL, NumHours, NumStudents, Revenue)

For each time slot, month, and city of the tennis club, show:

- Average revenue per student.
- Average number of lesson hours per student, separately by quarter.
- Ratio between the number of students and the total number of students, considering all tennis club located in the same province.

Perform the analyses separately by payment method.

```
SELECT TimeSlot , Month , City
SUM(Revenue)/SUM(NumStudents)
SUM(NumHours)/SUM(NumStudent)OVER(PARTITION BY 3-Months TimeSlot Month City)
SUM(NumStudents)/SUM(SUM(NumStudents)OVER(PARTITION BY Province TimeSlot Month City))
FROM TENNIS-LESSION T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI , TIME-HOUR TIH
WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL , TIH.IDTimeH = T.IDTimeH
GROUP BY TimeSlot , Month , City , PaymentMethod , 3-Months
```

Commento:

```
SELECT TimeSlot , Month , City
SUM(Revenue)/SUM(NumStudents)
SUM(SUM(NumHours)/SUM(SUM(NumStudent)OVER(PARTITION BY 3-Months TimeSlot Month City)
SUM(NumStudents)/SUM(SUM(NumStudents)OVER(PARTITION BY Province TimeSlot Month City))
FROM TENNIS-LESSION T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI , TIME-HOUR TIH
WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL , TIH.IDTimeH = T.IDTimeH
GROUP BY TimeSlot , Month , City , PaymentMethod , 3-Months, Province
```

Domanda 5

Risposta
corretta

Conceptual schema (1.6 points, -15% penalty for each wrong answer)

We want to analyze the information related to the use of university cafeterias in Italy.

The location of the cafeterias is known, expressed in terms of neighborhood, city, and region. Access to the cafeteria may be granted to different members of the university, i.e., students, faculty, and administrative staff. The cafeterias provide discount agreements on meal purchases, such as student discounts, off-peak discounts, and regional financial support, etc.

Customers who purchase meals at the cafeteria are characterized by their type of diet and the indication if they are students, faculty members or staff members; their age range is also known (one value between lower than 30 years, between 30 years and 50 years, greater than 50 years). In addition the customer reference address, expressed in terms of city, province region, and country, is stored.

We want to store the time the meal was purchased, expressed in terms of date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year. You also want to store the corresponding time slot (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm), and the payment method used by the customer (one value between "Cash", "Credit_Card", "DebitCard"). The purchased meal can include one or more courses between first course, main course, side dish and dessert.

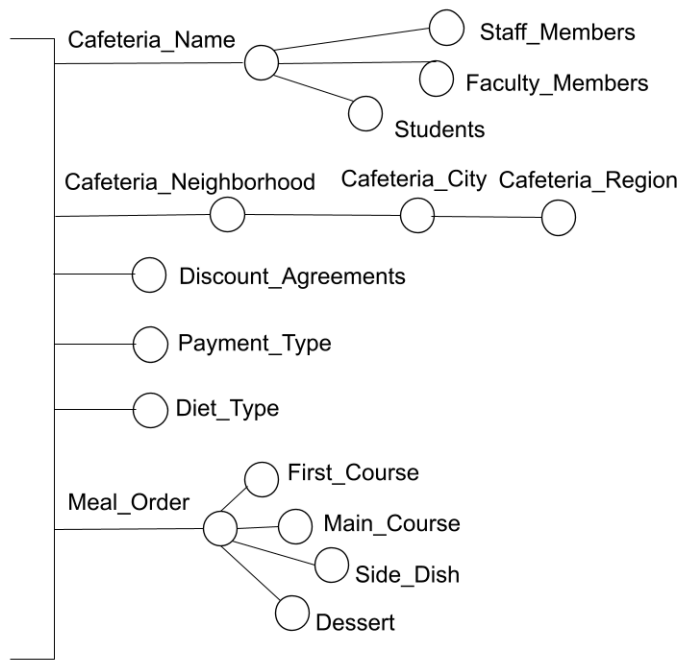
We want to analyze the average cost per customer and the average length of stay per customer based on the following information:

- Cafeteria, characterized by its name and its location expressed in terms of neighborhood, city, and region. The information about which members of the university can access the cafeteria is also available (one or more values between "Students", "Staff_Members", "Faculty_Members"). In addition, the list of discount agreements is available (e.g., student discount, off-peak discounts, regional financial support, etc.)
- Payment type (one value between "Cash", "Credit_Card", "DebitCard").
- Diet type (one value between "Regular", "Vegetarian", "Gluten_Free")
- Meal order (one or more values between "First_Course", "Main_Course", "Side_Dish", "Dessert")
- Customer reference address, expressed in terms of city, province region, and country
- Customer type (one value between "Students", "Staff_Members", "Faculty_Members")
- Customer age range (one value between lower than 30 years, between 30 years and 50 years, greater than 50 years)
- Date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year in which the customer purchased the meal; the corresponding time slot is known (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm).

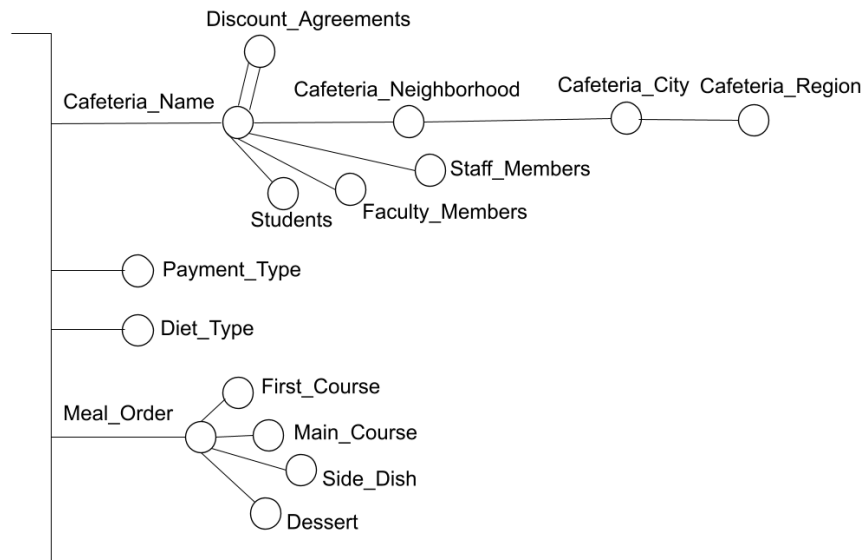
Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

- ☐ a. None of the other answers is correct

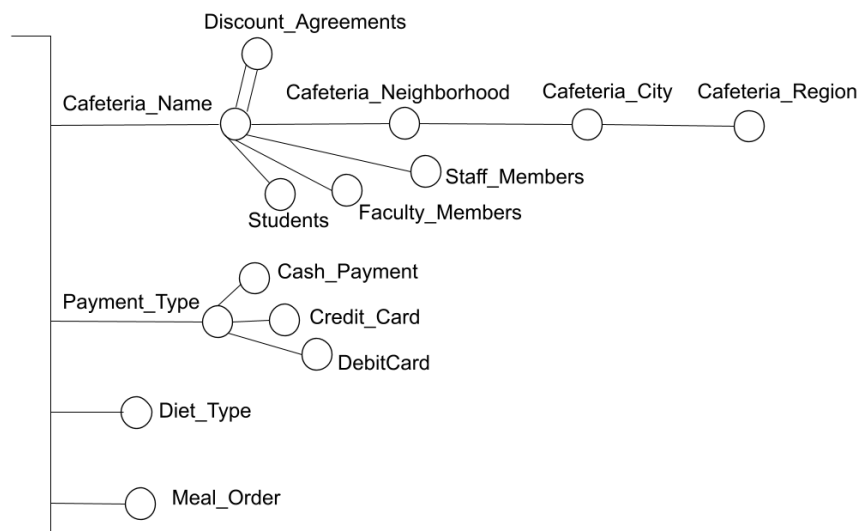
☐ b.



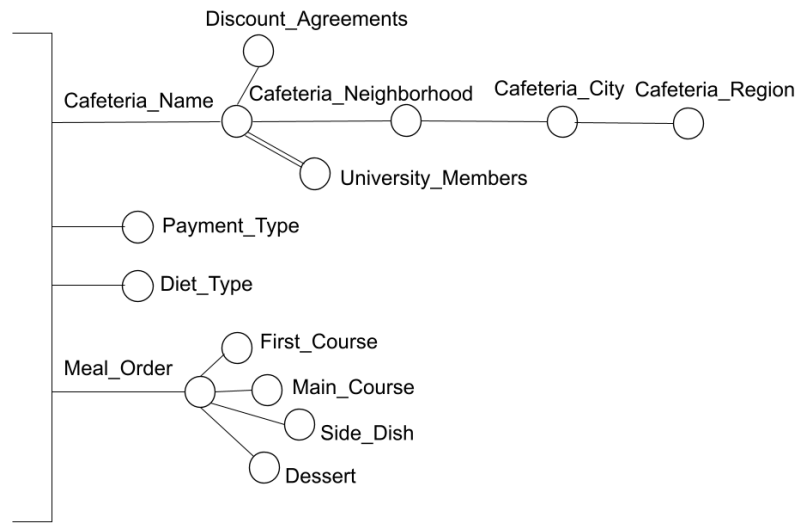
☒ c.



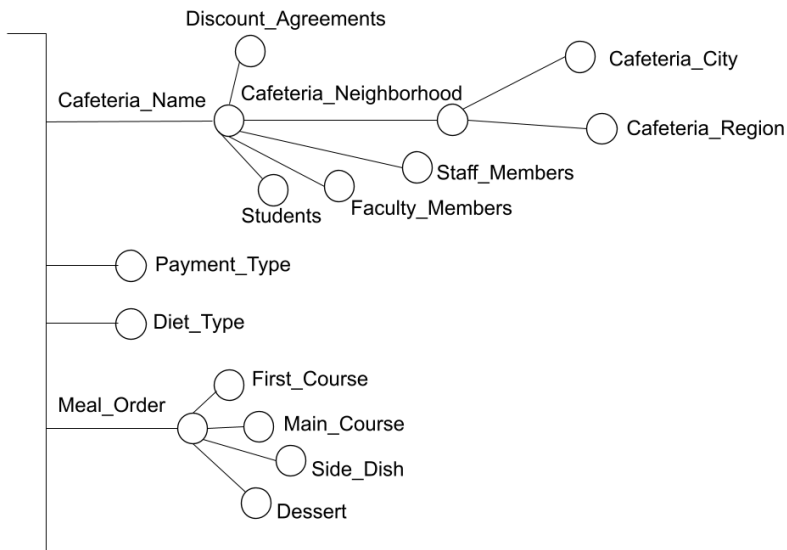
☐ d.



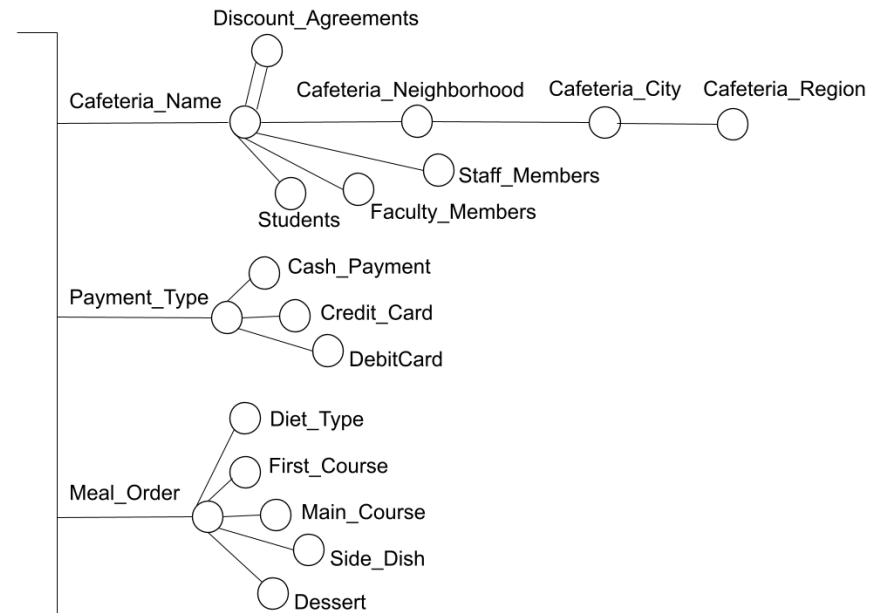
e.



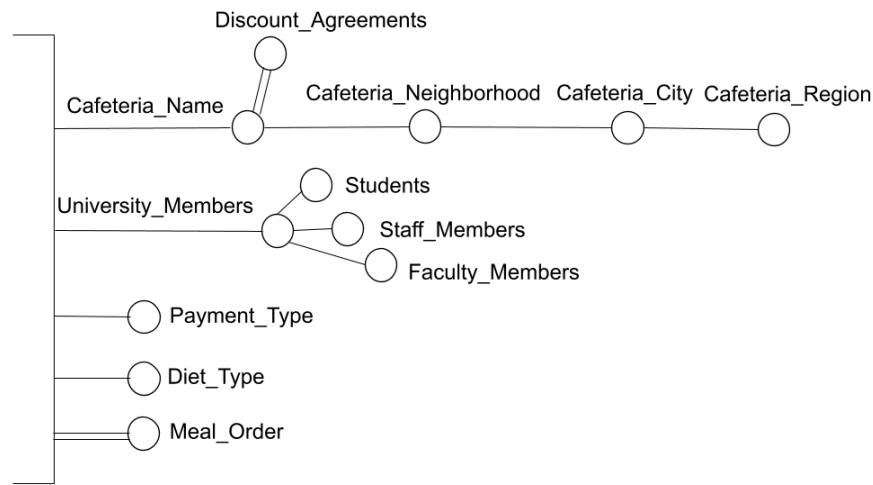
f.



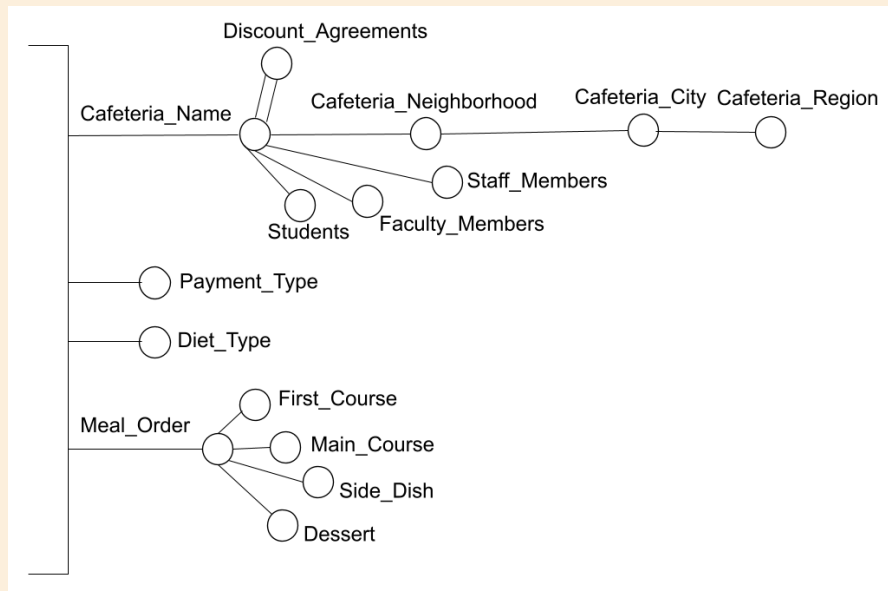
g.



h.



Risposta corretta.



La risposta corretta è:

Domanda 6

Risposta
errata

Punteggio
ottenuto -0,30
su 2,00

2 points (-15% penalty for each incorrect answer)

Consider the following transactional database:

TID	Items
1	{A, B, C, G}
2	{A, B, F, G, K}
3	{B, C, E, K}
4	{A, C, J, K}
5	{A, B, C, D}
6	{A, D, F, G}

Given a minimum support of 3 (i.e., an itemset is considered as frequent if it appears in at least 3 transactions), construct the corresponding FP-Tree. In the case of items with equal support, use lexicographical order (for example, if items A and B have the same support, then A precedes B).

What is the set of direct child nodes of the root and their respective counts, once the FP-Tree has been constructed? (The format is Node: Count)

- ☐ a. A: 1, B: 5
- ☐ b. None of the other answers is correct
- ☒ c. A: 5, B: 4, C: 4 ✖
- ☐ d. A: 5, B: 1, C: 1
- ☐ e. A: 6, B: 4, C: 4, G: 3, K: 3
- ☐ f. A: 5, B: 1
- ☐ g. A: 5, B: 4

Risposta errata.

La risposta corretta è:
A: 5, B: 1

Domanda 7

Parzialmente
corretta

Punteggio
ottenuto 0,67
su 1,00

Indexes (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
DOCTOR (DID, FirstName, LastName, Department, Salary)
PROTOCOL (PID, ProtocolName, TherapyType, MedicalArea, Duration)
EXECUTION (DID, PID, ExecutionDate)
PATIENT (PTID, FirstName, LastName, BirthDate, ResidenceCity)
ASSIGNMENT (PTID, PID, Outcome, AssignmentDate)
```

Assume the following cardinalities:

- $\text{card}(\text{DOCTOR}) = 2 \cdot 10^3$ tuples
 - Distinct values of the Department = 10
- $\text{card}(\text{PROTOCOL}) = 2 \cdot 10^4$ tuples
 - Distinct values of MedicalArea = 20
- $\text{card}(\text{EXECUTION}) = 4 \cdot 10^5$ tuples
 - $\text{MIN}(\text{ExecutionDate}) = 1/1/2006$, $\text{MAX}(\text{ExecutionDate}) = 31/12/2025$
- $\text{card}(\text{PATIENT}) = 10^5$ tuples
 - Distinct values of ResidenceCity = 200
 - $\text{MIN}(\text{BirthDate}) = 1/1/1950$, $\text{MAX}(\text{BirthDate}) = 31/12/2009$
- $\text{card}(\text{ASSIGNMENT}) = 4 \cdot 10^5$ tuple

Furthermore, assume the following reduction factor for the having clauses:

$\text{HAVING AVERAGE(D.Salary)} > 45000 = 1/4$

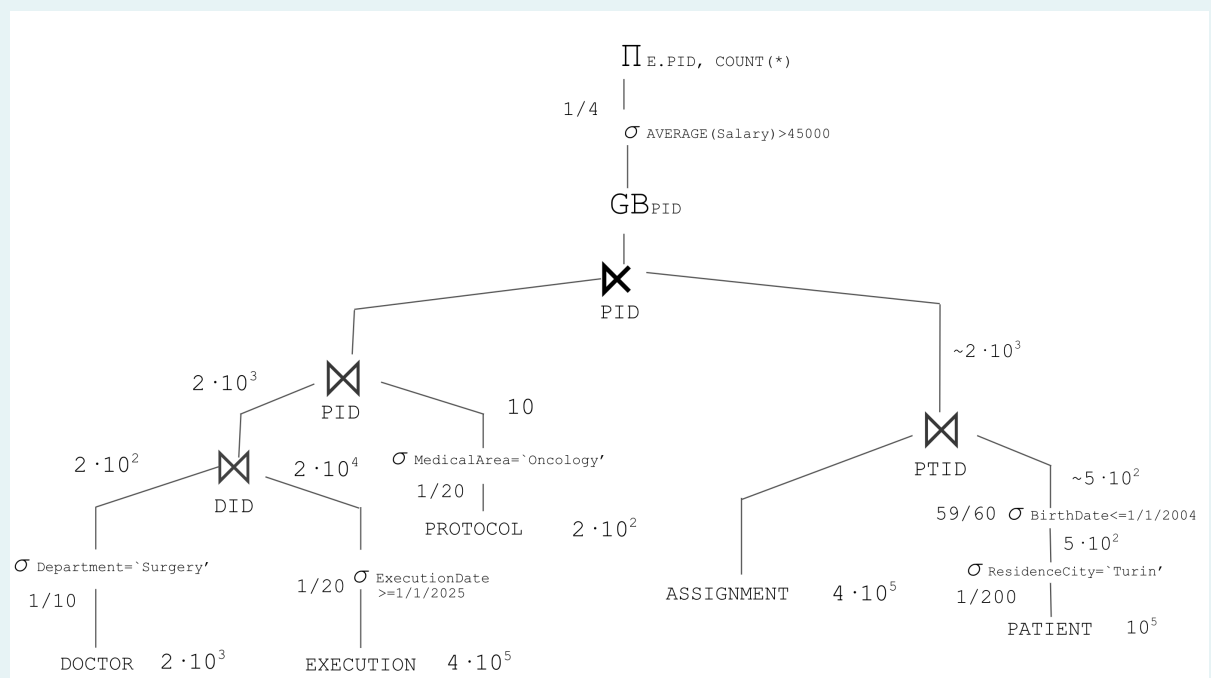
Consider the following query:

```

SELECT E.PID, COUNT(*)
FROM DOCTOR D, PROTOCOL P, EXECUTION E
WHERE D.DID = E.DID
      AND P.PID = E.PID
      AND P.MedicalArea = 'Oncology'
      AND E.ExecutionDate >= 1/1/2025
      AND D.Department = 'Surgery'
      AND E.PID IN (
        SELECT A.PID
        FROM ASSIGNMENT A, PATIENT PT
        WHERE A.PTID = PT.PTID
              AND PT.ResidenceCity = 'Turin'
              AND PT.BirthDate <= 1/1/2009
        )
GROUP BY E.PID
HAVING AVERAGE(D.Salary) > 45000;

```

The figure below represents the query tree for the query above.



Select one or more secondary physical structures to increase query performance (if possible) among the options below. You can select multiple correct answers.

- ☐ a. CREATE INDEX IndexJ ON PATIENT(ResidenceCity)- B+-Tree
- ☐ b. CREATE INDEX IndexA ON DOCTOR(Department)- HASH
- ☒ c. CREATE INDEX IndexD ON EXECUTION(ExecutionDate)- B+-Tree ✓
- ☐ d. CREATE INDEX IndexG ON PATIENT(BirthDate)- HASH
- ☐ e. CREATE INDEX IndexF ON PROTOCOL(MedicalArea)- B+-Tree
- ☐ f. CREATE INDEX IndexH ON PATIENT(BirthDate)- B+-Tree
- ☒ g. CREATE INDEX IndexI ON PATIENT(ResidenceCity)- HASH ✓
- ☐ h. None - the additional physical structures would not improve the query performance
- ☐ i. CREATE INDEX IndexC ON EXECUTION(ExecutionDate)- HASH
- ☐ j. CREATE INDEX IndexB ON DOCTOR(Department)- B+-Tree
- ☐ k. CREATE INDEX IndexE ON PROTOCOL(MedicalArea)- HASH

Risposta parzialmente esatta.

Hai selezionato correttamente 2.

Le risposte corrette sono: CREATE INDEX IndexA ON DOCTOR(Department)- HASH, CREATE INDEX IndexD ON EXECUTION(ExecutionDate)- B+-Tree, CREATE INDEX IndexI ON PATIENT(ResidenceCity)- HASH

Domanda 8

Parzialmente
corretta

Punteggio
ottenuto 0,03
su 1,00

Cardinalities (1 point, -15% penalty for each wrong answer)

The following tables are provided:

```
DOCTOR (DID, FirstName, LastName, Department, Salary)
PROTOCOL (PID, ProtocolName, TherapyType, MedicalArea, Duration)
EXECUTION (DID, PID, ExecutionDate)
PATIENT (PTID, FirstName, LastName, BirthDate, ResidenceCity)
ASSIGNMENT (PTID, PID, Outcome, AssignmentDate)
```

Assume the following cardinalities:

- $\text{card}(\text{DOCTOR}) = 2 \cdot 10^3$ tuples
 - Distinct values of the Department = 10
- $\text{card}(\text{PROTOCOL}) = 2 \cdot 10^4$ tuples
 - Distinct values of MedicalArea = 20
- $\text{card}(\text{EXECUTION}) = 4 \cdot 10^5$ tuples
 - $\text{MIN}(\text{ExecutionDate}) = 1/1/2006$, $\text{MAX}(\text{ExecutionDate}) = 31/12/2025$
- $\text{card}(\text{PATIENT}) = 10^5$ tuples
 - Distinct values of ResidenceCity = 200
 - $\text{MIN}(\text{BirthDate}) = 1/1/1950$, $\text{MAX}(\text{BirthDate}) = 31/12/2009$
- $\text{card}(\text{ASSIGNMENT}) = 4 \cdot 10^5$ tuples

Furthermore, assume the following reduction factor for the having clauses:

$\text{HAVING AVERAGE}(\text{D.Salary}) > 45000 = 1/4$

Consider the following query:

```
SELECT E.PID, COUNT(*)
FROM DOCTOR D, PROTOCOL P, EXECUTION E
WHERE D.DID = E.DID
      AND P.PID = E.PID
      AND P.MedicalArea = 'Oncology'
      AND E.ExecutionDate >= 1/1/2025
      AND D.Department = 'Surgery'
      AND E.PID IN (
        SELECT A.PID
        FROM ASSIGNMENT A, PATIENT PT
        WHERE A.PTID = PT.PTID
              AND PT.ResidenceCity = 'Turin'
              AND PT.BirthDate <= 1/1/2009
      )
GROUP BY E.PID
HAVING AVERAGE(D.Salary) > 45000;
```

The figure below represents the query tree for the query above.

- **lowercase:** all words are converted to lowercase
- **tokenization:** separation of the string into individual words that will be considered as comparison terms

Domanda 9

Risposta non
data

Punteggio
max.: 1,00

- **stemming**: reduction of the root form of each term (e.g., infinitive for verbs, masculine singular for adjectives, etc.)
- **no synonyms**: different words that mean the same thing are considered DIFFERENT

Query:

Distributed search engine

Documents:

- D1: A thorough search for faults should be distributed across all mechanical engine units.
- D2: Relational databases support transactions and structured queries.
- D3: A search engine retrieves relevant documents using an inverted index.
- D4: Distributed systems focus on scalability and fault tolerance.

According to ElasticSearch's full-text search, which document should have the highest relevance score? Assume that the distance between documents is calculated using only the term frequency (

$Tf(t \in D_i) = \sqrt{\text{frequency}}$) of the query words.

- ☐ a. D1
- ☐ b. D3
- ☐ c. It is not possible to answer the question with the available information.
- ☐ d. D4
- ☐ e. D2

Risposta errata.

La risposta corretta è:
D1

Domanda 10

Risposta non
data

Non valutata

This question is not a part of the exam

You can use the text area below to write any note or draft (e.g. intermediate steps of an exercise).

Any text written below will not be considered toward the correction of the exam.

**Domanda
11**

Risposta
corretta

Punteggio
ottenuto 1,60
su 1,60

Conceptual schema (1.6 points, -15% penalty for each wrong answer)

We want to analyze the information related to the use of university cafeterias in Italy.

The location of the cafeterias is known, expressed in terms of neighborhood, city, and region. Access to the cafeteria may be granted to different members of the university, i.e., students, faculty, and administrative staff. The cafeterias provide discount agreements on meal purchases, such as student discounts, off-peak discounts, and regional financial support, etc.

Customers who purchase meals at the cafeteria are characterized by their type of diet and the indication if they are students, faculty members or staff members; their age range is also known (one value between lower than 30 years, between 30 years and 50 years, greater than 50 years). In addition the customer reference address, expressed in terms of city, province region, and country, is stored.

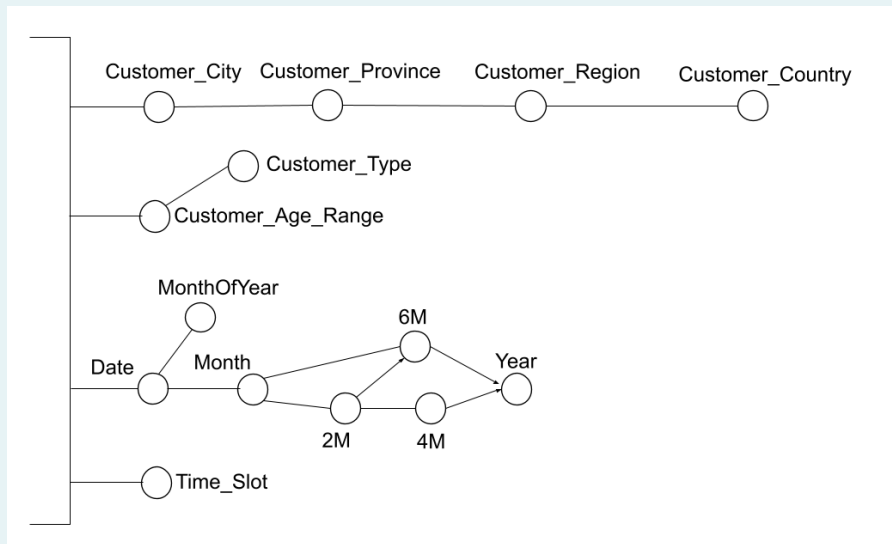
We want to store the time the meal was purchased, expressed in terms of date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year. You also want to store the corresponding time slot (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm), and the payment method used by the customer (one value between "Cash", "Credit_Card", "DebitCard"). The purchased meal can include one or more courses between first course, main course, side dish and dessert.

We want to analyze the average cost per customer and the average length of stay per customer based on the following information:

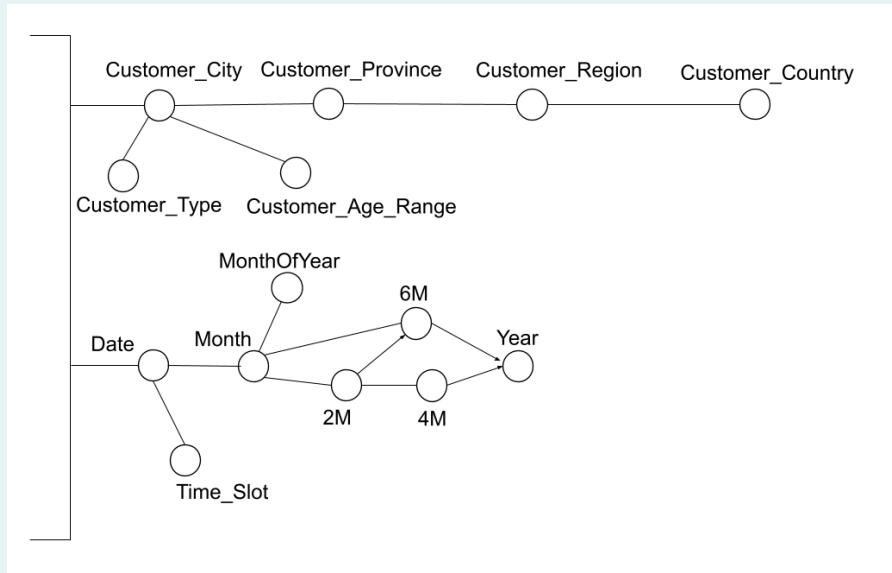
- Cafeteria, characterized by its name and its location expressed in terms of neighborhood, city, and region. The information about which members of the university can access the cafeteria is also available (one or more values between "Students", "Staff_Members", "Faculty_Members"). In addition, the list of discount agreements is available (e.g., student discount, off-peak discounts, regional financial support, etc.)
- Payment type (one value between "Cash", "Credit_Card", "DebitCard").
- Diet type (one value between "Regular", "Vegetarian", "Gluten_Free")
- Meal order (one or more values between "First_Course", "Main_Course", "Side_Dish", "Dessert")
- Customer reference address, expressed in terms of city, province region, and country
- Customer type (one value between "Students", "Staff_Members", "Faculty_Members")
- Customer age range (one value between lower than 30 years, between 30 years and 50 years, greater than 50 years)
- Date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year in which the customer purchased the meal; the corresponding time slot is known (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm).

Select, from the dimensions proposed below, those that meet the requirements described in the problem specifications.

☐ a.

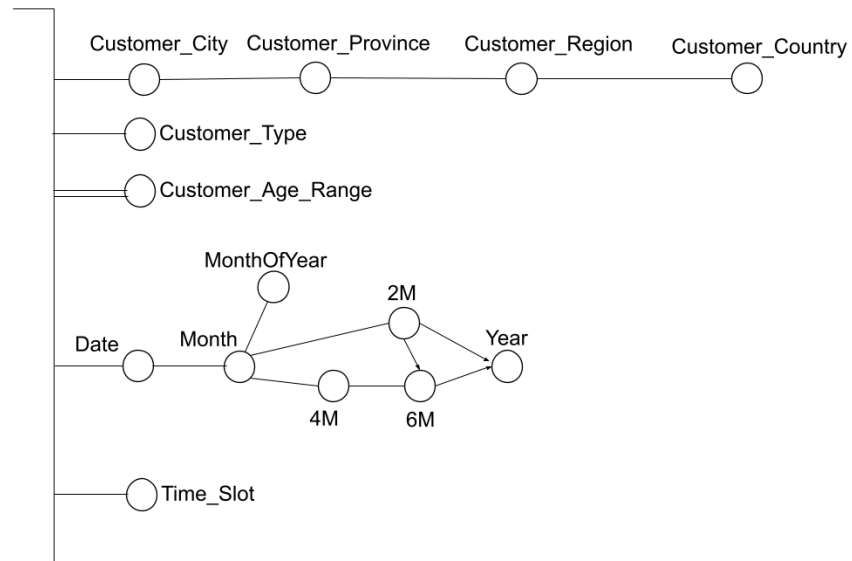


☐ b.

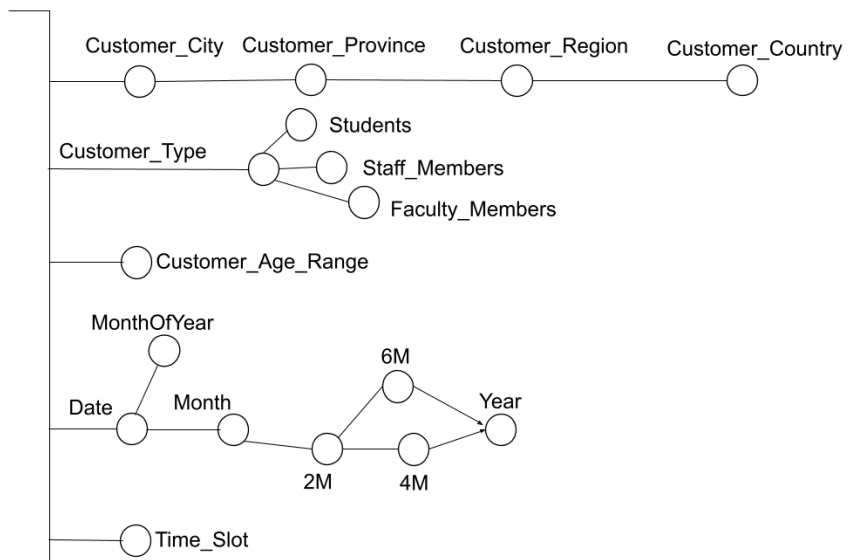


☐ c. None of the other answers is correct

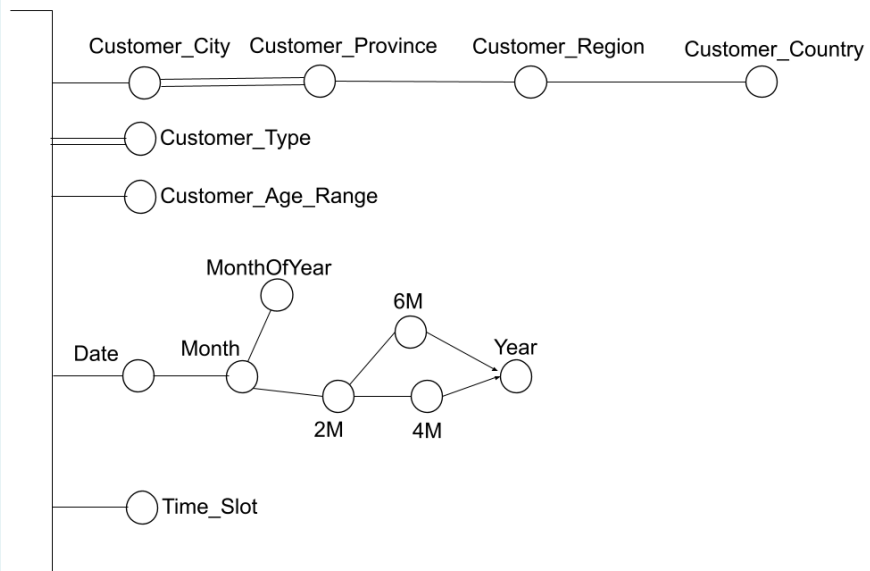
d.



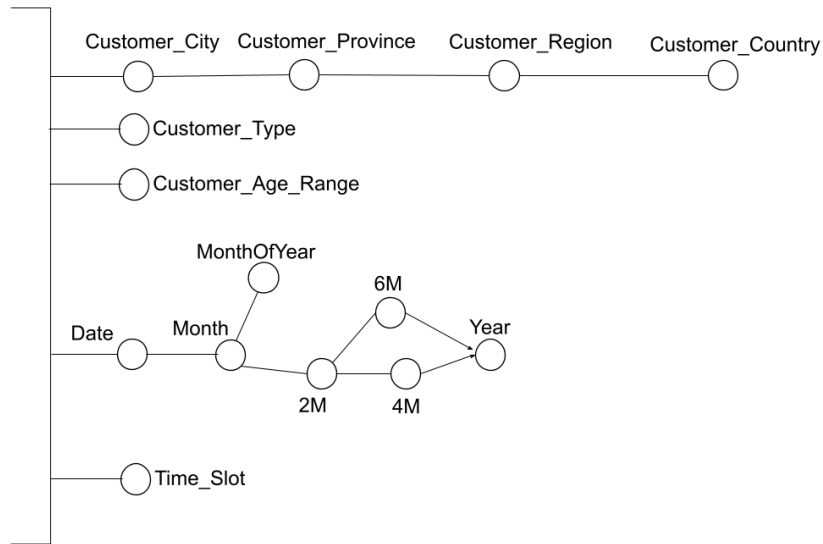
e.



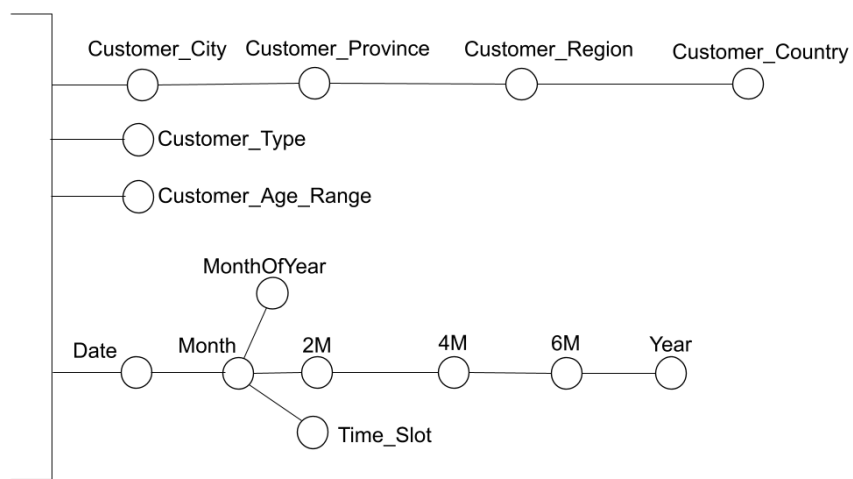
f.



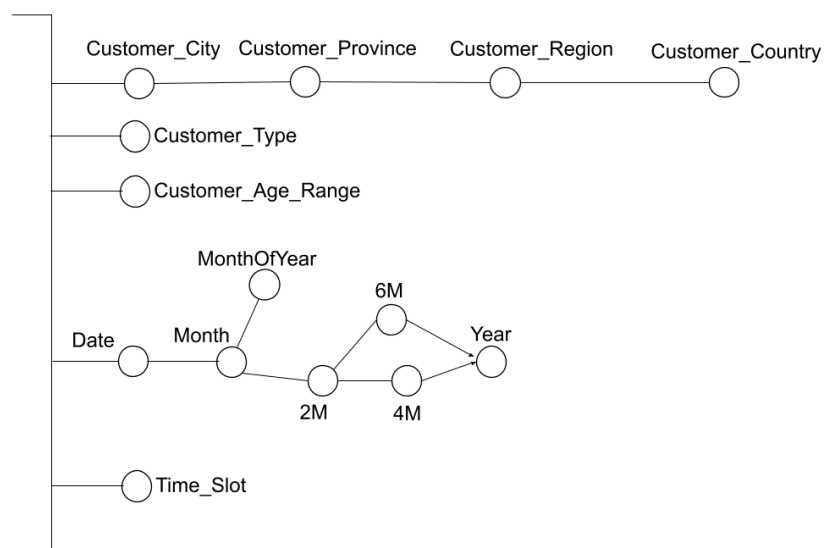
☒ g.



☐ h.



Risposta corretta.



La risposta corretta è:

1 point (-15% penalty for each incorrect answer)

Domanda 12

Risposta errata

Punteggio ottenuto -0,15 su 1,00

Consider a hierarchical agglomerative clustering dendrogram built on an 8-point dataset using Complete Linkage (MAX).

During the clustering process, mergers occur at the following distances (heights in the dendrogram):

- First merger at distance 0.2
- Second merger at distance 0.4
- Third merger at distance 1.4
- Fourth merger at distance 2.0
- Final merger at distance 3.5

If the dendrogram is cut at height (i.e. distance) 1.0, how many clusters are obtained?

- ☐ a. 3 clusters
- ☐ b. 4 clusters
- ☐ c. None of the other answers is correct
- ☒ d. 2 clusters ✖
- ☐ e. 5 clusters
- ☐ f. 6 clusters

Risposta errata.

La risposta corretta è:
5 clusters

Domanda 13

Completo

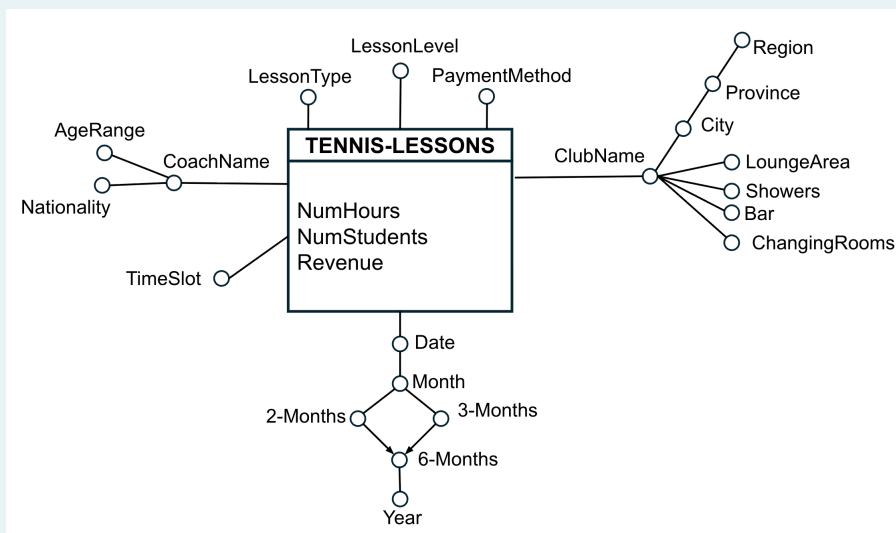
Punteggio ottenuto 1,00 su 4,00

Extended SQL (4 points)

The following data warehouse stores information related to the management of tennis lessons for a chain of tennis clubs located throughout Italy. We want to analyze, on a daily basis and by time slot (morning from 9:00 a.m. to 1:00 p.m., afternoon from 1:00 p.m. to 5:00 p.m., and evening from 5:00 p.m. to 9:00 p.m.), the number of hours, the number of students, and the amount of revenue collected.

The tennis clubs are characterized by a geographical hierarchy and by the available services, such as the presence of a lounge area, bar, showers, and changing rooms (configuration attribute). Tennis coaches are characterized by nationality and age range. The analysis also considers the lesson type (individual or group), the lesson level (beginner/advanced/competition), and the payment method (cash/card).

The data warehouse is characterized by the following conceptual schema and the corresponding logical schema.



TENNIS-CLUB (IDClub, ClubName, City, Province, Region, LoungeArea, Showers, Bar, ChangingRooms)

COACH (IDCoach, CoachName, Nationality, AgeRange)

TIME-DATE (IDTimeD, Date, Month, 2-Months, 3-Months, 6-Months, Year)

TIME-HOUR (IDTimeH, TimeSlot)

JUNK-LESSON (IDJL, LessonType, LessonLevel, PaymentMethod)

TENNIS-LESSONS (IDClub, IDCoach, IDTimeD, IDTimeH, IDJL, NumHours, NumStudents, Revenue)

For each coach, lesson level, and for each semester, show:

- average number of lesson hours per month
- total number of lesson hours, separately by year
- percentage of lesson hours with respect to the total lesson hours delivered by all instructors of the same nationality

Assign a ranking position for each record, calculated based on the total number of lesson hours, separately by nationality

```
SELECT CouchName LessonLevel 3-Months
```

```
SUM(NumHours)/SUM(SUM(Numhours))OVER(PARTION BY Month CouchName LessonLevel 3-Month)
```

```
SUM(SUM(NumHours))OVER(PARTION BY Year couchName LessonLevel 3-Month)
```

```
SUM(SUM(Numhours))/SUM(SUM(Numhours))OVER(PARTION BY Nationality)
```

```
RANK()OVER(PARTION BY Nationality CouchName LessonLevel ORDER BY NumHours DESC)
```

```
FROM TENNIS-LESSON T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI , COACH C
```

```
WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL AND C.IDCoach =T.IDCouch
```

```
GROUOP BY CouchName LessonLevel 3-Months Month Nationality
```

Commento:

SELECT CouchName LessonLevel 3-Months

SUM(NumHours)/~~SUM(SUM(Numhours))OVER(PARTION BY Month CouchName LessonLevel 3-Month)~~ COUNT(DISTINCT Month)

SUM(SUM(NumHours))OVER(PARTION BY Year couchName LessonLevel ~~3-Month~~)

~~SUM(SUM(Numhours))~~/SUM(SUM(Numhours))OVER(PARTION BY Nationality, LessonLevel, 6M)

RANK()~~OVER(PARTION BY Nationality CouchName LessonLevel~~ORDER BY SUM(NumHours) DESC)

FROM TENNIS-LESSON T , JUNK-LESSON J , TENNIS-CLUB TE , TIME-DATE TI , COACH C

WHERE T.IDClub = TE.IDCub AND T.IDTimeD = TI.IDTimeD AND J.IDJL = TIDJL AND C.IDCoach =T.IDCouch

GROUOP BY CouchName LessonLevel ~~36-Months~~ ~~Month~~ Nationality, Year

**Domanda
14**

Risposta
corretta

Punteggio
ottenuto 1,80
su 1,80

Measures (1.8 points, -15% penalty for each wrong answer)

We want to analyze the information related to the use of university cafeterias in Italy.

The location of the cafeterias is known, expressed in terms of neighborhood, city, and region. Access to the cafeteria may be granted to different members of the university, i.e., students, faculty, and administrative staff. The cafeterias provide discount agreements on meal purchases, such as student discounts, off-peak discounts, and regional financial support, etc.

Customers who purchase meals at the cafeteria are characterized by their type of diet and the indication if they are students, faculty members or staff members; their age range is also known (one value between lower that 30 years, between 30 years and 50 years, greater than 50 years). In addition the customer reference address, expressed in terms of city, province region, and country, is stored.

We want to store the time the meal was purchased, expressed in terms of date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year. You also want to store the corresponding time slot (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm), and the payment method used by the customer (one value between "Cash", "Credit_Card", "DebitCard"). The purchased meal can include one or more courses between first course, main course, side dish and dessert.

We want to analyze the average cost per customer and the average length of stay per customer based on the following information:

- Cafeteria, characterized by its name and its location expressed in terms of neighborhood, city, and region. The information about which members of the university can access the cafeteria is also available (one or more values between "Students", "Staff_Members", "Faculty_Members"). In addition, the list of discount agreements is available (e.g., student discount, off-peak discounts, regional financial support, etc.)
- Payment type (one value between "Cash", "Credit_Card", "DebitCard").
- Diet type (one value between "Regular", "Vegetarian", "Gluten_Free")
- Meal order (one or more values between "First_Course", "Main_Course", "Side_Dish", "Dessert")
- Customer reference address, expressed in terms of city, province region, and country
- Customer type (one value between "Students", "Staff_Members", "Faculty_Members")
- Customer age range (one value between lower that 30 years, between 30 years and 50 years, greater than 50 years)
- Date, month, month of the year, 2-month period (2M), four-month period (4M), six-month period (6M) and year in which the customer purchased the meal; the corresponding time slot is known (one value between before 12 pm, between 12 pm and 2pm, between 2 pm and 3 pm).

Select from the list only those attributes necessary to correctly model the measures in the fact table required by the specifications.

- ☐ a. Average length of stay per customer, Average cost per customer
- ☐ b. Average length of stay per customer, Total number of cafeterias, Average cost per customer
- ☒ c. Total length of stay, Total number of customers, Total cost ✓
- ☐ d. Average cost per cafeteria, Total length of stay, Total number of customers
- ☐ e. Maximum cost, Minimum length of stay, Total number of customers
- ☐ f. Total number of meal types, Total number of diets, Total cost
- ☐ g. Average length of stay per customer, Total number of customers, Average cost per customer
- ☐ h. None of the other answers is correct
- ☐ i. Total cost, Total length of stay, Total number of cafeterias

Risposta corretta.

La risposta corretta è:

Total length of stay, Total number of customers, Total cost

**Domanda
15**

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-15% for incorrect answer)

Documentation

- find - Retrieves documents that meet certain conditions, allowing you to select certain fields as 'projection'.
Syntax: `db.collection.find(<query>, <projection>)`
- \$gt - (Greater Than) Selects documents where the value of a field is greater than a specified value.
Syntax: `{ field: { $gt: value } }`
- \$elemMatch - Allows you to apply multiple conditions to the same element in an array.
Syntax: `{ array_field: { $elemMatch: { condition1, condition2 } } }`
- \$or - Selects documents that meet at least one of several conditions.
Syntax:

```
{ $or: [
  { condition1 },
  { condition2 },
  { condition3 }
]}
```

Consider a MongoDB collection called tracks, which represents the catalogue of a music streaming platform. Here is an example document:

```
{
  "_id": ObjectId("64f0c9a8e1b2c3d4e5f6a7b8"),
  "title": "Midnight Echoes",
  "genres": ["Indie Pop", "Synthwave"],
  "release_date": "2023-05-20",
  "artist": {
    "name": "Luna Ray",
    "country": "Italy"
  },
  "daily_statistics": [
    {
      "date": "2024-01-15",
      "country": "Germany",
      "number_of_streams": 15000,
      "average_duration_in_seconds": 180
    },
    {
      "date": "2024-01-16",
      "country": "Italy",
      "number_of_streams": 25000,
      "average_duration_in_seconds": 210
    }
  ],
  "label": "Neon Records",
  "explicit_content": false
}
```

Write a MongoDB query to find all songs that meet the following conditions:

- They were released after 1 January 2023;
- They are by artists from Italy;
- They have at least one record in daily_statistics with more than 20,000 streams or an average duration of more than 200 seconds.

The query should only return the song title and artist name.

☒ a.

```
db.tracks.find(
  {
    release_date: { $gt: "2023-01-01" },
    "artist.country": "Italy",
    daily_statistics: {
      $elemMatch: {
        $or: [
          { number_of_streams: { $gt: 20000 } },
          { average_duration_in_seconds: { $gt: 200 } }
        ]
      }
    }
  },
  {
    _id: 0,
    title: 1,
    "artist.name": 1
  }
)
```

 ✓

☐ b.

```
db.tracks.find(
  {
    release_date: { $gt: "2023-01-01" },
    "artist.country": "Italy",
    daily_statistics: {
      $elemMatch: {
        $or: [
          { number_of_streams: { $gt: 20000 } },
          { average_duration_in_seconds: { $gt: 200 } }
        ]
      }
    }
  },
)
```

☐ c.

```
db.tracks.find(
  {
    daily_statistics: {
      $elemMatch: {
        release_date: { $gt: "2023-01-01" },
        "artist.country": "Italy",
        number_of_streams: { $gt: 20000 },
        average_duration_in_seconds: { $gt: 200 }
      }
    }
  },
)
```

☐ d.

```
db.tracks.find(
  {
    release_date: { $gt: "2023-01-01" },
    "artist.country": "Italy",
    daily_statistics: {
      $elemMatch: {
        number_of_streams: { $gt: 20000 },
        average_duration_in_seconds: { $gt: 200 }
      }
    }
  },
)
```

☐ e.

```
db.tracks.find(
  {
    daily_statistics: {
      $elemMatch: {
        release_date: { $gt: "2023-01-01" },
        "artist.country": "Italy",
        number_of_streams: { $gt: 20000 },
        average_duration_in_seconds: { $gt: 200 }
      }
    }
  },
  {
    _id: 0,
    title: 1,
    "artist.name": 1
  }
)
```

☐ f.

```
db.tracks.find(
  {
    release_date: { $gt: "2023-01-01" },
    "artist.country": "Italy",
    daily_statistics: {
      $elemMatch: {
        number_of_streams: { $gt: 20000 },
        average_duration_in_seconds: { $gt: 200 }
      }
    }
  },
  {
    _id: 0,
    title: 1,
    "artist.name": 1
  }
)
```

☐ g. None of the other answers is correct

Risposta corretta.

La risposta corretta è:

```
db.tracks.find(
  {
    release_date: { $gt: "2023-01-01" },
    "artist.country": "Italy",
    daily_statistics: {
      $elemMatch: {
        $or: [
          { number_of_streams: { $gt: 20000 } },
          { average_duration_in_seconds: { $gt: 200 } }
        ]
      }
    }
  },
  {
    _id: 0,
    title: 1,
    "artist.name": 1
  }
)
```

**Domanda
16**

Completo

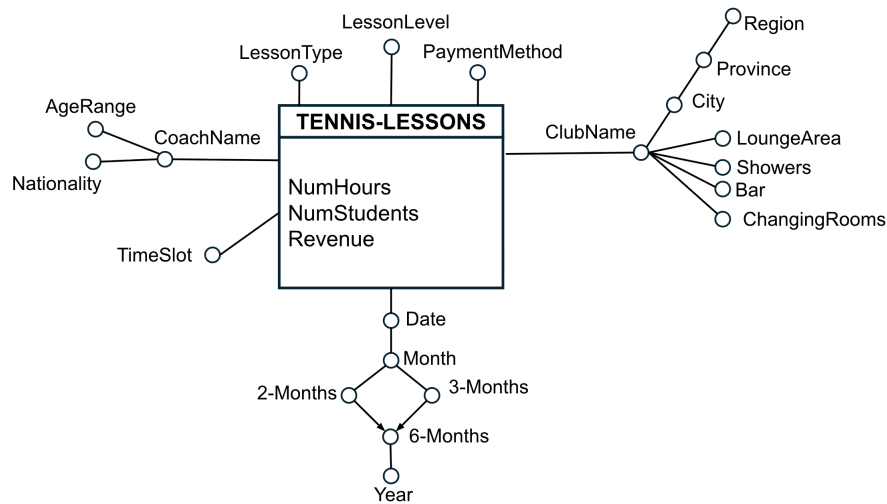
Punteggio
ottenuto 0,00
su 5,00

Materialized view (5 points)

The following data warehouse stores information related to the management of tennis lessons for a chain of tennis clubs located throughout Italy. We want to analyze, on a daily basis and by time slot (morning from 9:00 a.m. to 1:00 p.m., afternoon from 1:00 p.m. to 5:00 p.m., and evening from 5:00 p.m. to 9:00 p.m.), the number of hours, the number of students, and the amount of revenue collected.

The tennis clubs are characterized by a geographical hierarchy and by the available services, such as the presence of a lounge area, bar, showers, and changing rooms (configuration attribute). Tennis coaches are characterized by nationality and age range. The analysis also considers the lesson type (individual or group), the lesson level (beginner/advanced/competition), and the payment method (cash/card).

The data warehouse is characterized by the following conceptual schema and the corresponding logical schema.



TENNIS-CLUB (IDClub, ClubName, City, Province, Region, LoungeArea, Showers, Bar, ChangingRooms)

COACH (IDCoach, CoachName, Nationality, AgeRange)

TIME-DATE (IDTimeD, Date, Month, 2-Months, 3-Months, 6-Months, Year)

TIME-HOUR (IDTimeH, TimeSlot)

JUNK-LESSON (IDJL, LessonType, LessonLevel, PaymentMethod)

TENNIS-LESSONS (IDClub, IDCoach, IDTimeD, IDTimeH, IDJL, NumHours, NumStudents, Revenue)

Given the above logical scheme, consider the following queries of interest:

- For clubs with showers and changing rooms but no bar, display the cumulative value of lesson revenue for each quarter (3-Month attribute), separately by year, payment method (PaymentMethod attribute) and club region.
- Separately for each two-month period, lesson type and lesson level, show the percentage of the revenue compared to the total revenue per year and the average number of students per lesson hour, considering only clubs in the city of Turin.
- For "group" lessons (LessonType attribute), show the total revenue and average revenue per student separately by year and club province.

Given the above logical schema, answer the following requests:

- Define a materialized view with the CREATE MATERIALIZED VIEW, that can be used to efficiently answer all three of the above queries (i.e., a, b, c). Specifically, define the query in SQL associated with BLOCK A in the following instruction

```

CREATE MATERIALIZED VIEW ViewLessons
BUILD IMMEDIATE
REFRESH FAST ON COMMIT
AS
Block A
  
```

- Define the **minimal set** of attributes that allows identifying the tuples belonging to the materialized view ViewLessons.
- Assume that the management of the materialized view (derived table) is carried out by means of triggers. Write the trigger to propagate to the ViewLessons materialized view the changes due to the insertion of a new record into the TENNIS-LESSONS fact table

```

SELECT ClubName Showers ChargingRoom
SUM(SUM(Revenue))OVER(PARTION BY Year ReORDER BY 3-Month )
  
```

Commento:

SELECT ClubName Showers ChargingRoom

SUM(SUM(Revenue))OVER(PARTITION BY Year ReORDER BY 3-Month)

The solution is incomplete

**Domanda
17**

Risposta non
data

Punteggio
max.: 1,50

1.5 points (-15% penalty for each incorrect answer)

Let us consider a dataset with 3 classes C_1, C_2, C_3

The a priori probabilities are:

- $P(C_1) = 0.4$
- $P(C_2) = 0.35$
- $P(C_3) = 0.15$
- $P(C_4) = 0.1$

There is a binary attribute ($X \in \{\text{Yes}, \text{No}\}$), with conditional probabilities:

Class $P(X = \text{Yes} \mid C)$ $P(X = \text{No} \mid C)$

C_1	0.2	0.8
C_2	0.6	0.4
C_3	0.8	0.2
C_4	0.7	0.3

A new instance has $X = \text{Yes}$.

According to the Naïve Bayes classifier, to which class is the instance assigned?

- ☐ a. There is not enough information to determine the answer.
- ☐ b. C_1
- ☐ c. All classes are equally probable.
- ☐ d. C_4
- ☐ e. C_3
- ☐ f. C_2

Risposta errata.

La risposta corretta è:

C_2